Parthiv Patel

+1 825-343-5376 | Edmonton, Alberta | parthiv@ualberta.ca | LinkedIn: parthiv06 | Portfolio →

Education

Bachelor of Science – Mechatronics and Robotics Engineering Co-op **Class of 2029**
University of Alberta - GPA: 3.5/4.0 *Edmonton, AB*

- **Coursework:** Object-Oriented Programming, Digital Logic Design, Circuit Design, Deformable Bodies
- **Awards:** Dean's Research Award, First Class Standing, Opportunities Award, Jason Lang Scholarship

Technical Skills

Embedded Systems: Arduino, Raspberry Pi, ESP32, Microcontroller Interfacing (I^2C, SPI, UART)
Electronics & PCB Design: KiCAD, Altium, PCB layout, Schematic capture, SMT Soldering
Mechanical Design: SolidWorks, Fusion 360, AutoCAD
Programming: Python, C/C++, MATLAB, Linux Environment, JavaScript, VBA
Manufacturing & Prototyping: 3D Printing, CNC Milling, CNC Lathe, Laser Cutting

Work Experience

Mechanical Engineering Summer Student **May 2025 – August 2025**
Universe Machine Corporation *Edmonton, AB*

- Designed valve components by taking shop-floor measurements, producing AutoCAD drawings, and applying design for manufacturing (DFM) and Geometric Dimensioning and Tolerancing (GD&T).
- Performed stress concentration, pin strength, and pressure design calculations to design valve modifications.
- Digitized and indexed 30+ years of engineering drawings to improve retrieval efficiency and support the transition to digital authentication under APEGA standards.
- Automated workflows using Python scripts and Excel VBA macros for CSV data processing increasing efficiency by 50%.

Production Team Lead **January – June 2024**
Foldify – A Junior Achievement Company *Edmonton, AB*

- Co-led a 20-member team in design, production, and sales of 100+ clothes-folding boards, generating $2,500 in revenue.
- Oversaw the design process, iterating multiple versions before finalizing a three-flap folding board for practical home use, and coordinated manufacturing for large-scale production.
- Engaged with customers and stakeholders at 10+ trade shows and business events, demonstrating strong interpersonal, presentation, and relationship-building skills.

Project Experience

Powertrain Team Member | UAlberta Formula SAE **September 2025 – Present**

- Designed a PCB using KiCAD to enable power switching between the 650V tractive battery and 12V low-voltage circuit.
- Collaborating with a team to develop embedded Teensy ECU firmware for a CAN-controlled inverter supporting regenerative braking, using GitHub for version control and adhering to Formula SAE Electric rules.
- Tuned and tested the inverter EEPROM settings using a dynamometer and performed efficiency calculations comparing recovered energy to thermal losses with a peak of 16 Amps reverse current during regenerative braking.

Electro-mechanical Team Member | UAlberta Aerial Robotics Group **September 2025 – Present**

- Fabricated PCBs using stencil-applied solder paste, surface-mount component placement, and reflow soldering.
- Used SolidWorks to model electronic component-mounting brackets with appropriate tolerancing, using FEA for validation, followed by 3D printing and integration.
- Designed and prototyped a servo-actuated tilt mechanism for dual motors, including carbon fiber rod mounting, motor mounts, and friction-fit assemblies, ensuring mechanical strength under full propulsion.

Drone Error-Display Module | Embedded Systems Personal Project **November 2025 – March 2026**

- Designed Raspberry Pi HAT PCB in KiCad integrating I^2C, SPI, and UART interfaces for drone error monitoring.
- Assembled the PCBA and integrated ESP32 firmware for communication and display control.

Robotic Gripper | Mechatronics Personal Project **December 2025 – January 2026**

- Designed and manufactured a geared robotic gripper using Fusion 360 and additive manufacturing.
- Integrated stepper motor actuation using an Arduino controller and ULN2003 driver.